LSI at BAREC Shared Task 2026: A Controlled Study of Training Convergence, Heuristics, and Ensembles for Arabic Sentence-Level Readability

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Abstract

We present the Linguistic Security Institute (LSI) submission to the BAREC Shared Task 2026 (Task 1): predicting Arabic sentence readability on a 19-level ordinal scale. Our approach uses a logit-level ensemble of three Arabic pre-trained language models, with CAMeLBERT trained using an ordinal Earth Mover's Distance loss and augmented by six SAMER Lexicon difficulty features. Controlled comparisons show that surface heuristics improve undertrained baselines but degrade converged models, and retrieval-augmented in-context learning yields no gains in this fine-grained setting. Our final ensemble achieves 81.7 QWK, ranking 1st among Constrained-track system submissions and 3rd among Open-track system submissions.

1 Introduction

Arabic readability assessment is challenging due to rich morphology, register variation, diglossia, and limited fine-grained labeled data (Al Khalil et al., 2017). The BAREC corpus (Elmadani et al., 2025; Habash et al., 2025) addresses this gap with 69,441 sentences annotated on a 19-level ordinal scale across three domains and three readership groups, enabling detailed sentence-level modeling of difficulty. We participate in both tracks: Constrained (BAREC plus SAMER resources) and Open.

We study three complementary Arabic pre-trained language models: AraBERT (Antoun et al., 2020), CAMeLBERT-Mix (Inoue et al., 2021) (trained on a mixture of Modern Standard Arabic, Dialectal Arabic, and Classical Arabic), and AraModernBERT (Elshehy et al., 2026), and combine them in a logit-level ensemble. We report two empirical lessons for fine-grained ordinal classification. First, simple surface heuristics can help undertrained checkpoints but degrade converged models. Second, retrieval-augmented in-context learning does not yield gains for 19-level ordinal readability in our setup.

All experiments ran on consumer hardware and free cloud GPUs. We contribute a controlled analysis of heuristic effects, a negative result for RAG-ICL, and a strong ensemble system.

2 Background

2.1 The BAREC Challenge

BAREC-ST-2026 (Elmadani et al., 2026) includes multiple subtasks for Arabic readability assessment. This paper focuses on **Task 1: Sentence-Level Readability Assessment**, where models must predict one of 19 difficulty levels (1 = easiest, 19 = hardest) for each sentence. BAREC does not contain intentional dialectal variation: some sentences or words may resemble dialectal Arabic, but none of the BAREC data sources is intended to contain dialectal data. The Constrained track allows only BAREC training data plus the SAMER Corpus (Alhafni et al., 2025) and SAMER Lexicon (Al Khalil et al., 2020). The Open track permits any external resources. We participate in both tracks.

2.2 Readability Assessment Across Languages

Early Arabic readability work relied on surface cues such as word and sentence length (Al Khalil et al., 2017), later incorporating morphological and syntactic features with machine learning classifiers (Liberato et al., 2024). The SAMER project contributed foundational resources, including a leveled reading corpus and a word-difficulty lexicon that we use as explicit lexical features in our system (Al Khalil et al., 2018, 2020). While these early efforts targeted coarse 3–5 level schemes, the BAREC corpus extends this to 19 fine-grained levels, demanding sensitivity to much subtler lexical and syntactic differences.

Parallel work in English readability has followed a similar trajectory. Handcrafted linguistic features

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tend to provide the greatest benefit when training data or model capacity are limited, while fully converged transformers capture many readability cues implicitly (Vajjala and Meurers, 2012; Lee et al., 2021; Deutsch et al., 2020). Indeed, Martinc et al. (2021) showed that fine-tuned BERT models consistently outperform feature-based approaches, establishing neural methods as the state of the art. For second-language readability, ranking-based formulations have also been shown to generalize better than direct classification (Xia et al., 2016).

More recently, retrieval-augmented in-context learning (RAG-ICL) has shown promise for Arabic binary classification tasks (Djamai et al., 2026). However, we find that this success does not transfer to fine-grained ordinal readability: retrieved neighbors are often topically similar but span multiple difficulty levels, introducing label noise rather than clear pedagogical signals. This observation aligns with findings from other languages, where fine-grained readability benchmarks have similarly highlighted the difficulty of obtaining reliable multi-level annotations across diverse genres (Vásquez-Rodríguez et al., 2022). Our work extends these cross-linguistic insights by systematically comparing heuristic and neural approaches under controlled convergence conditions.

3 System Overview

We study three pre-trained language models (PLMs) fine-tuned for readability prediction, chosen for complementary training data and architectures: AraBERT (Antoun et al., 2020) (tweet-heavy, informal), AraModernBERT (Elshehy et al., 2026) (large-scale modern Arabic with long context), and CAMeLBERT-Mix (Inoue et al., 2021) (trained on a mixture of Modern Standard Arabic, Dialectal Arabic, and Classical Arabic). This diversity helps the ensemble cover register and domain variation in BAREC; BAREC itself does not contain intentional dialectal data (though some sentences/words may resemble dialectal Arabic).

Constrained Track. We use only the BAREC training data (54,845 sentences) and the SAMER Lexicon (Al Khalil et al., 2020). From the lexicon, we extract six sentence-level features: (1) average difficulty, (2) percentage of hard words (≥ 4), (3) percentage of very hard words ($= 5$), (4) maximum difficulty, (5) standard deviation, and (6) lexicon coverage. We selected these features based on prior work showing that word-level difficulty correlates strongly with text readability (Al Khalil et al., 2020). We attach these features to the `[CLS]` embedding before classification to supply explicit lexical difficulty cues. Lexicon coverage is 12% without MADAMIRA lemmatization (Pasha et al., 2014); this low coverage reflects that the lexicon stores uninflected lemmas while the sentence surface strings contain prefixes and suffixes that prevent direct string matching.

For CAMeLBERT, we replace standard cross-entropy with Earth Mover's Distance (EMD) loss (Hou et al., 2016) to explicitly penalize ordinal distance:

$$\mathcal{L}_{\text{EMD}}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{K-1} \sum_{k=1}^{K-1} \left| \sum_{j=1}^{k} y_j - \sum_{j=1}^{k} \hat{y}_j \right|, \qquad K = 19. \tag{1}$$

Here $\mathbf{y}$ is the one-hot gold label vector and $\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$ is the predicted distribution. The loss compares cumulative distributions, so predictions that place probability mass farther from the gold level receive a larger penalty. For example, confusing easy vs. hard levels is penalized much more than an off-by-one error. This aligns EMD with the ordinal structure that QWK is designed to capture.

Open Track. Our Open Track submission uses the same ensemble and features as the Constrained Track, without any additional external labeled data.

Table 1 shows our controlled experiments across four AraBERT versions. V1 (CPU, 1 epoch, batch size 2) is undertrained and serves as our baseline. V2 adds simple rules: sentences with more than 20 words get +1 level; fewer than 5 words get -1; STEM and Specialized texts get +1; Foundational texts get -1. These rules improve QWK by +2.45 but collapse exact accuracy to 14.56%, confirming they help only undertrained models.

V3 (GPU, 4 epochs, early stopping at epoch 2, batch size 16, max length 256) achieves the best raw performance with 74.39 QWK and the lowest MAE (1.44). We used early stopping with patience 2 on development QWK, as preliminary experiments showed performance plateaued after epoch 2 on GPU.

Crucially, V4 applies the same post-processing rules to V3 and illustrates why we treat QWK, exact Accuracy, and MAE as complementary signals: the rules slightly increase QWK (74.39 $\rightarrow$ 74.69) by nudging borderline predictions closer to the gold level (fewer large off-by-many errors), but they also break exact label alignment (46.47% $\rightarrow$ 29.59%)

Ver.	HW	Ep	Bs	Len	Loss	QWK	Acc	MAE
V1	CPU	1	2	128	CE	68.7	47.3	1.5
V2	CPU+h	1	2	128	CE	71.1	14.6	2.2
V3	GPU	4	16	256	CE	**74.4**	46.5	**1.4**
V4	GPU+h	4	16	256	CE	74.7	29.6	1.7

Table 1: AraBERT progression. V3 is the converged raw model used in the final ensemble. h = heuristics.

and worsen overall calibrated distance (MAE 1.44 $\rightarrow$ 1.68). This makes the "heuristic inversion" explicit: once a checkpoint is well-trained, hard rules can marginally improve an ordinal distance metric while degrading exact-match and average error.

We also tested these rules on AraModernBERT; they hurt performance, so we used the raw AraModernBERT in both ensembles.

We build two ensembles with equal weights (0.34/0.33/0.33). Ensemble 1 (based on V2): AraBERT V2 + AraModernBERT + CAMeLBERT, yielding 78.84 development QWK and 81.5 blind QWK. Ensemble 2 (based on V3): AraBERT V3 + AraModernBERT + CAMeLBERT, yielding 78.03 development QWK and **81.7 blind QWK**. Despite lower development scores, Ensemble 2 generalizes better (+3.67 vs +2.66 QWK gain). This confirms that raw converged logits are more robust than heuristic-adjusted ones.

4 Experimental Setup

All models were trained on the official BAREC training split (54,845 sentences, 79.0%), with hyperparameter selection, checkpoint saving, and early stopping conducted on the fixed development set (7,310 sentences, 10.5%). The remaining 10.5% (7,286 sentences) formed the blind test set. Model evaluation used the official BAREC `eval.py` script across all 19 difficulty classes.

4.1 Evaluation Metric

We use Quadratic Weighted Kappa (QWK) (Cohen, 1968), which rewards ordinal agreement and penalizes mistakes by squared distance:

$$QWK = 1 - \frac{\sum_{i=1}^{K}\sum_{j=1}^{K} w_{ij}O_{ij}}{\sum_{i=1}^{K}\sum_{j=1}^{K} w_{ij}E_{ij}}, \qquad w_{ij} = \frac{(i-j)^2}{(K-1)^2}, \quad K = 19. \tag{2}$$

Here O_{ij} is the observed confusion matrix, E_{ij} is the expected matrix under chance agreement, and w_{ij} assigns larger penalties to errors farther from the gold level.

Setting	AraBERT V3	AraModernBERT	CAMeLBERT
Learning rate	2e-5	2e-5	2e-5
Batch size	16	2	16
Epochs	4 (early stop)	1	2
Max length	256	128	128
Loss	CE	CE	EMD
Hardware	Kaggle T4	CPU i5	Kaggle T4

Table 2: Key hyperparameters (full training configuration in the repository).

Text sequences were tokenized using model-specific subword tokenizers with truncation up to maximum context lengths. Transformer architectures were fine-tuned end-to-end using AdamW with linear learning rate warmup over the initial 10% of steps followed by linear decay. Cross-entropy models (AraBERT V3 and AraModernBERT) used default targets. CAMeLBERT used Earth Mover's Distance (EMD) loss with an ordinal cost matrix to penalize logit distance. Early stopping monitored development QWK (patience = 2).

AraBERT V3 and CAMeLBERT V2 were fine-tuned on single 16GB Kaggle NVIDIA T4 GPUs (< 4 hours each). Memory constraints required training AraModernBERT on CPU (Intel Core i5-8300H @ 2.30GHz, 16GB RAM), completing in 37 hours. Table 2 summarizes configurations.

5 Results

We present development results in Table 3 and official blind test results in Table 4. Overall, broader pretraining strengthens single-model performance, heuristics help undertrained models but can harm converged ones, and ensembling improves robustness. The development set analyses directly informed our model selection and ensemble design. The blind test results confirm that these choices generalize to unseen data. On the blind test, our final ensemble achieves 81.7 QWK.

5.1 Development Set Performance

Development set performance (Table 3) shows AraModernBERT leading single-model performance (75.53 QWK), followed by AraBERT V3 (74.39) and CAMeLBERT with SAMER features (73.73). Ensemble 1, which uses heuristic-adjusted AraBERT outputs, achieves higher development QWK than Ensemble 2 (78.84 vs 78.03), but Ensemble 2, which averages raw converged logits, generalizes better on unobserved data.

The **heuristic inversion** is visible when comparing V1 vs V2 and V3 vs V4: heuristics raise QWK for the undertrained V1 (+2.45) but reduce

#	Model	QWK	Acc	Acc±1	MAE	Acc7	Acc5	Acc3
1	AraBERT V1	68.7	47.3	61.2	1.5	56.1	61.2	68.3
2	AraBERT V2	71.1	14.6	42.3	2.2	32.2	49.9	71.4
3	AraBERT V3	74.4	46.5	—	1.4	—	—	—
4	AraBERT V4	74.7	29.6	—	1.7	—	—	—
5	AraModernBERT	75.5	54.1	67.7	1.3	63.0	67.1	73.9
6	CAMeLBERT V2	73.7	46.2	60.9	1.5	57.0	62.8	71.2
7	Ensemble 1	78.8	28.3	64.9	1.5	48.9	61.2	74.1
8	Ensemble 2	78.0	27.0	61.3	1.6	46.1	59.1	74.1

Table 3: Development set results (%). Ensemble 2 is our final submission. Acc7/5/3 report collapsed accuracies after mapping 19 levels to 7, 5, and 3 bins; — indicates metrics not logged in early experiments.

Track	Rank	QWK	Acc	Acc±1	MAE	Acc7	Acc5	Acc3
Baseline	—	82.5	42.4	73.6	1.1	61.6	68.8	75.6
Constrained	**#1**	**81.7**	50.6	72.9	**1.0**	65.0	71.2	77.2
Open	**#3**	**81.7**	50.6	72.9	**1.0**	65.0	71.2	77.2

Table 4: Official blind test evaluation results (%). The same ensemble submission was entered in both tracks.

exact accuracy for the converged V3 (46.47% to 29.59%).

5.2 Blind Test Performance

On the official blind test set (Table 4), Ensemble 2 achieves 81.7 QWK, ranking 1st among Constrained-track system submissions and 3rd among Open-track system submissions. Table 4 also reports the official baseline (82.5 QWK, 42.4% accuracy, 1.1 MAE). The demo baseline reports higher exact accuracy (55.8% vs 50.6%) (Altarbouch et al., 2025).

6 Discussion

Heuristic utility depends heavily on model convergence. While surface rules boost undertrained baselines (+2.45 QWK), they over-correct converged models and severely degrade exact accuracy (V3 $\rightarrow$ V4: -16.88 percentage points) by overriding learned syntactic and lexical signals.

Ensembling three diverse PLMs strengthens performance on the blind test: averaging model outputs smooths extreme misclassifications and yields our lowest MAE (1.0). RAG-ICL (`multilingual-e5-large`, FAISS, $k = 5$) did not help: retrieved neighbors matched the domain but varied widely in difficulty (3–5 levels), injecting label noise rather than useful demonstrations. This highlights that, unlike many binary settings, topical similarity is not a reliable signal for fine-grained ordinal readability.

Limitations and Reproducibility. Our study has several methodological and computational constraints. First, SAMER Lexicon feature extraction relied on surface string matching without morphological normalization (e.g., stemming or lemmatization), restricting lexical coverage to 12%. Second, hardware limitations constrained AraModernBERT to CPU training with a small batch size, preventing extensive hyperparameter tuning across model architectures. Third, our ensemble relied on equal, uncalibrated logit weights rather than learned stacking. Finally, our negative findings on RAG-ICL reflect a fixed retrieval configuration (`multilingual-e5-large`, $k = 5$) and may not generalize across retrieval frameworks or index strategies.

7 Conclusion

We presented the LSI submission for BAREC Task 1: an ensemble of three Arabic PLMs augmented with SAMER Lexicon features and trained with Earth Mover's Distance. On the official blind test set, our approach achieved 81.7 QWK, ranking 1st among Constrained-track system submissions and 3rd among Open-track system submissions.

Beyond leaderboard results, our controlled experiments highlight two takeaways for fine-grained ordinal prediction. (1) Surface heuristics exhibit an inversion effect: they can inflate distance-sensitive metrics on undertrained checkpoints, but degrade exact accuracy and increase error after convergence. (2) RAG-ICL was ineffective under our setting, as topically similar neighbors often differed by multiple readability levels and introduced label noise.

For more information, visit our GitHub repository: https://github.com/LinguisticSecurityInstitute/BAREC-2026.

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